Teaching notes

In this activity, students make simple 'organisms' from different materials. Working in pairs, they use the same material to make a population of organisms by following very simple rules.

When they have made their population they study each individual and note down any differences between them. This reveals variation within the population.
Students look at the differences and try to decide what environmental change these particular individuals may be better adapted to.

Keywords

Adaptation, offspring, population, reproduce, species, survive, variation (continuous and discontinuous variation).

The idea is to demonstrate how variation within a population may help individuals of the same species survive when the environment changes.
As an extension it can be used to illustrate continuous and discontinuous variation.

Activity

1. Choose an organism from the following suggestions for students to make. Give them a certain amount of time to make their population e.g. 60 seconds. Challenge students to make them as identical as possible.

Organism 1

Work in pairs (to represent parents and shows that variation occurs even amongst offspring who share the same parents).
Using modelling dough, students make as many pea-sized balls of dough as they can in the given time.

Organism 2

Cut out very simple winged shapes from paper strips folded in half. The shapes should be no larger than 2 cm^2 when opened out.

Organism 3

Using the back of a pair of scissors, paper-curl (pre-cut 1cm wide strips from width of A4 sheet) paper strips.

2. Next, they decide broadly which habitat their organisms live in.

- Are they herbivores, carnivores or omnivores?
- Are they a prey species for other organisms?
- Are they camouflaged?
- Are they nocturnal?
- Do they hibernate?

When they have their population they carefully study each individual and make a note of any differences such as size, shape, smoothness, colour. Can they group them?
Now, they decide an environmental change which might favour the more unusual individuals.

E.g. Small pea shapes may survive better should there be a shortage of food in their habitat; darker peas may have an advantage if the climate becomes wetter and more cloudy.

Extension - continuous and discontinuous variation

Students decide which forms of variation in their population can be described and measured by using numbers and units and which can only be described using words and the number in each category tallied.

They can collate their data and share with other pairs who made the same creature.
Bar charts and histograms can be drawn.

Discuss the differences in bar charts and histograms e.g. bars are not touching on the chart because the categories are quite separate and distinct (discontinuous variation) whereas a histogram shows continuous variation and the bars are touching.

Discuss how charts can be used to show which part/proportion of a population shown on the graph may survive to pass on genes to its offspring, should a particular environmental change occur.

Discuss the effects of rapid changes in climate on species in evolutionary terms. There may not be sufficient variation in a population to allow it to adapt to a decrease in precipitation, for example.

To stimulate discussion, there are video clips available showing penguins which breed on the rocky shores of South Africa. Due to rising sea temperatures the fish stocks that they feed on are moving further away from the shore making harder for the penguins to feed. The population has declined by 60% since 2004.

www.youtube.com/watch?v=f9CIeuOfYXE